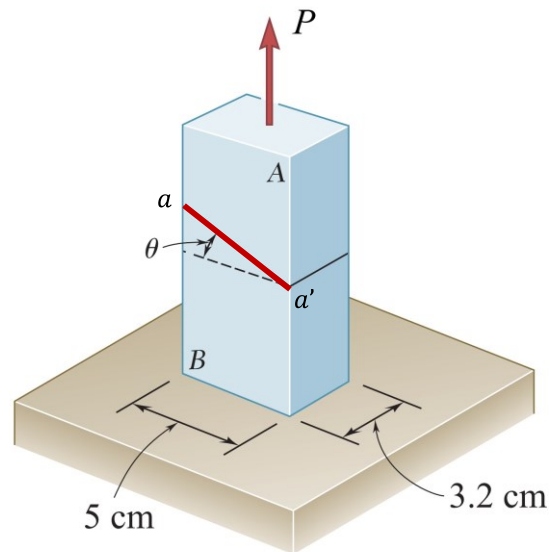


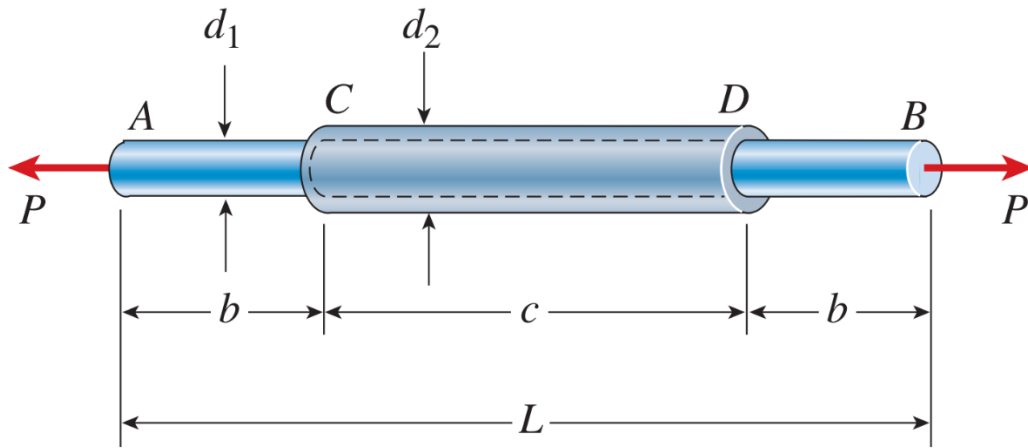
Final Exam for Solid Mechanics (EMCH260 Section 002, Fall 2021)
9:00 – 11:30 AM, Wednesday, Dec. 8th 2021

Name (First/Last): _____ Score: _____/100'

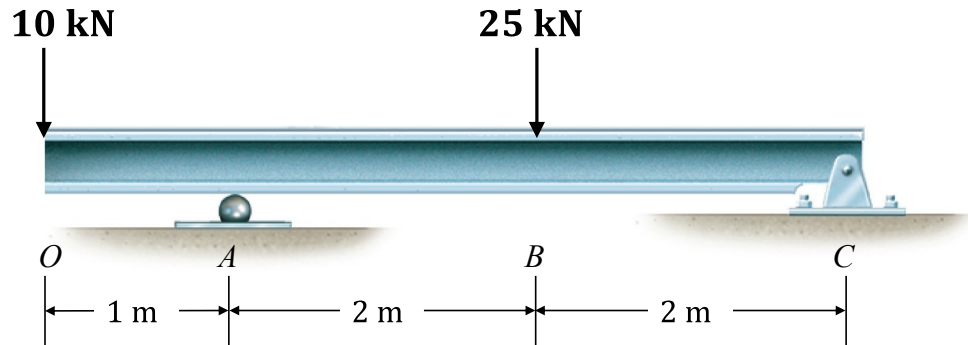
1. The two portions of member *A* and *B* are glued together along a plane *aa'* forming an angle $\theta = 30^\circ$ with respect to the horizontal. The lateral size of the member is $5\text{ cm} \times 3.2\text{ cm}$. Knowing that the allowable stress for the glued joint is 50 MPa in tension and 25 MPa in shear, determine the maximum possible external force (*P*) so that the glued joint will not break. (10' in total)



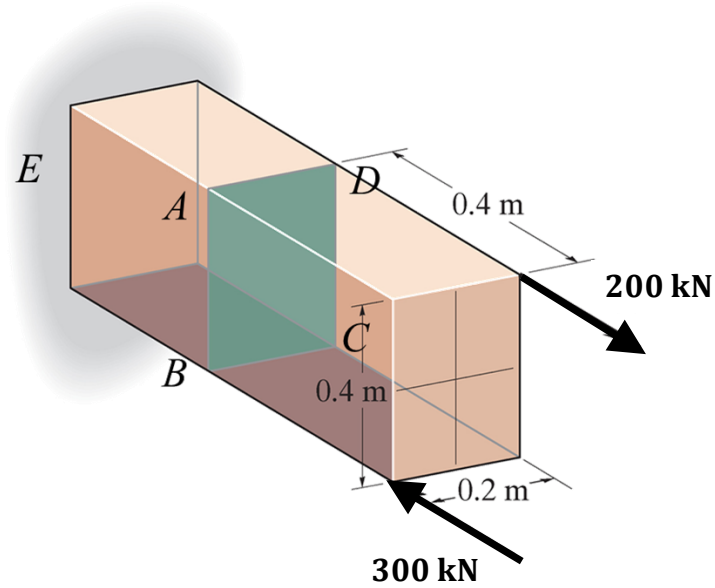
2. A plastic rod AB of length $L = 0.5$ m has a diameter $d_1 = 30$ mm (see figure). A plastic sleeve CD of length $c = 0.3$ m and outer diameter $d_2 = 45$ mm is securely bonded to the rod so that no slippage can occur between the rod and the sleeve ($b = 0.1$ m). The rod is made of an acrylic with modulus of elasticity $E_1 = 3.1$ GPa and Poisson's ratio $\nu = 0.3$, and the sleeve is made of a polyamide with $E_2 = 2.5$ GPa. Determine the elongation (δ) of the rod AB and the new diameter (d_1') of the AC or DB segment, when the rod is pulled by axial forces $P = 12$ kN. Note: take $\pi = 3.1415926$. (15' in total)



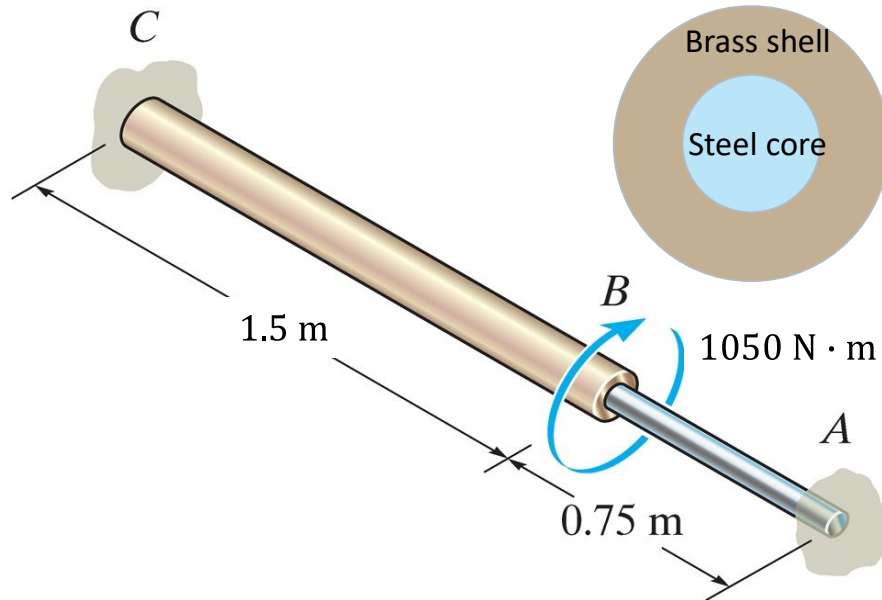
3. Determine the shear and bending moment diagrams of the beam with external loading shown below, and also determine the absolute maximum normal stress in the beam. The beam has a uniform square cross-section with lateral size $a = 0.1$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (15' in total)



4. The brick is subjected to the loads shown. The cross section throughout the entire brick is $0.2\text{ m} \times 0.4\text{ m}$. Determine the stress at the corner point A , B , C , and D . The cross section $ABCD$ is perpendicular to the longitudinal direction of the brick. The end E of the rod is fixed. (10' in total)

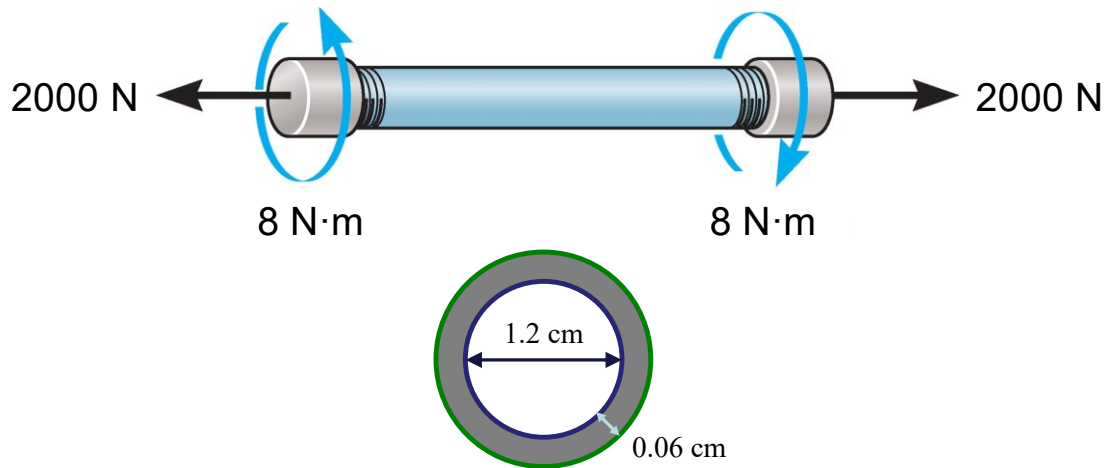


5. A shaft is made from two segments: AB is **pure steel** and BC is **core-shell** (core: steel, shell: brass). The entire shaft is fixed at its two ends and subjected to a torque of $T = 1050 \text{ N} \cdot \text{m}$ at point B . The AB portion has a diameter of 20 mm. The inner diameter and outer diameter of the core-shell part is 20 mm and 40 mm, respectively. The shear modulus of elasticity of steel and brass is 80 GPa and 40 GPa, respectively. Determine the reactions at the two ends A and C . (15' in total)

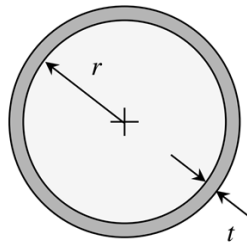
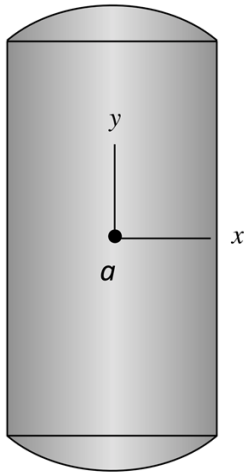


6. The thin-walled pipe has an inner diameter of 1.2 cm and a thickness of 0.06 cm. If it is subjected to an internal pressure of 4 MPa and the axial tension and torsional loadings shown, determine the principal stress and the absolute maximum shear stress for a point on the surface of the pipe. (15' in total)

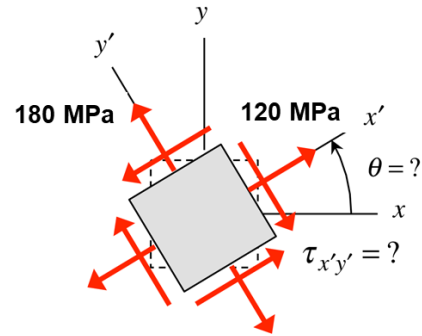
Hint: the cross-section of the vessel can be regarded as a hollow tube and the cross-section area can be calculated as $\frac{\pi}{4}(0.0132^2 - 0.012^2)$. You can do the same treatment for the polar moment inertia.



7. The closed thin-walled tank has an inner radius of r and wall thickness t and contains a gas of pressure of p . The state of stress at point “a” on the surface of the tank is represented by stress components with an unknown stress element rotation angle of θ . Note that, the two normal stress components for $x'y'$ system (note: not xy system!) are known: $\sigma_{x'} = 120$ MPa and $\sigma_{y'} = 180$ MPa, but the shear stress component ($\tau_{x'y'}$) is unknown. Determine: (20' in total)
- (1) the principal stresses at point a ; (10')
 - (2) $\tau_{x'y'}$; (5')
 - (3) rotation angle of θ . (5')

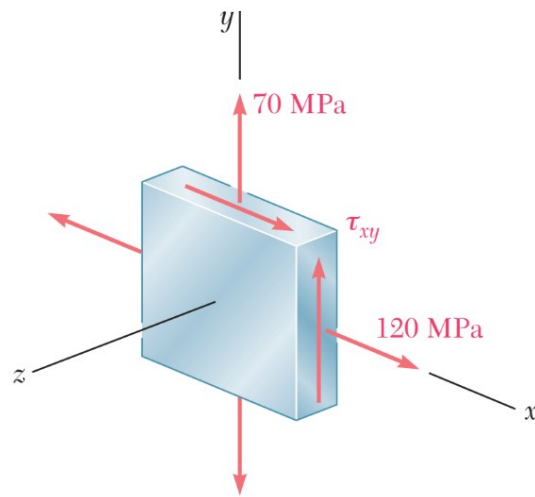


tank cross section



stress state at point a

***[**Bonus Problem 1**] For a plane stress state shown in the figure, if the absolute maximum shear stress is 130 MPa, determine the possible principal stresses in the xy -plane. (10' in total)



***[**Bonus Problem 2**] A pressurized steel tank has inner radius $r = 0.5$ m and wall thickness $t = 20$ mm. The steel has modulus of elasticity $E = 260$ GPa and Poisson's ratio $\nu = 0.3$. For a point D on the surface of the pressurized tank, if the absolute maximum shear strain is 5×10^{-4} , determine the internal pressure of the tank. (10' in total)

